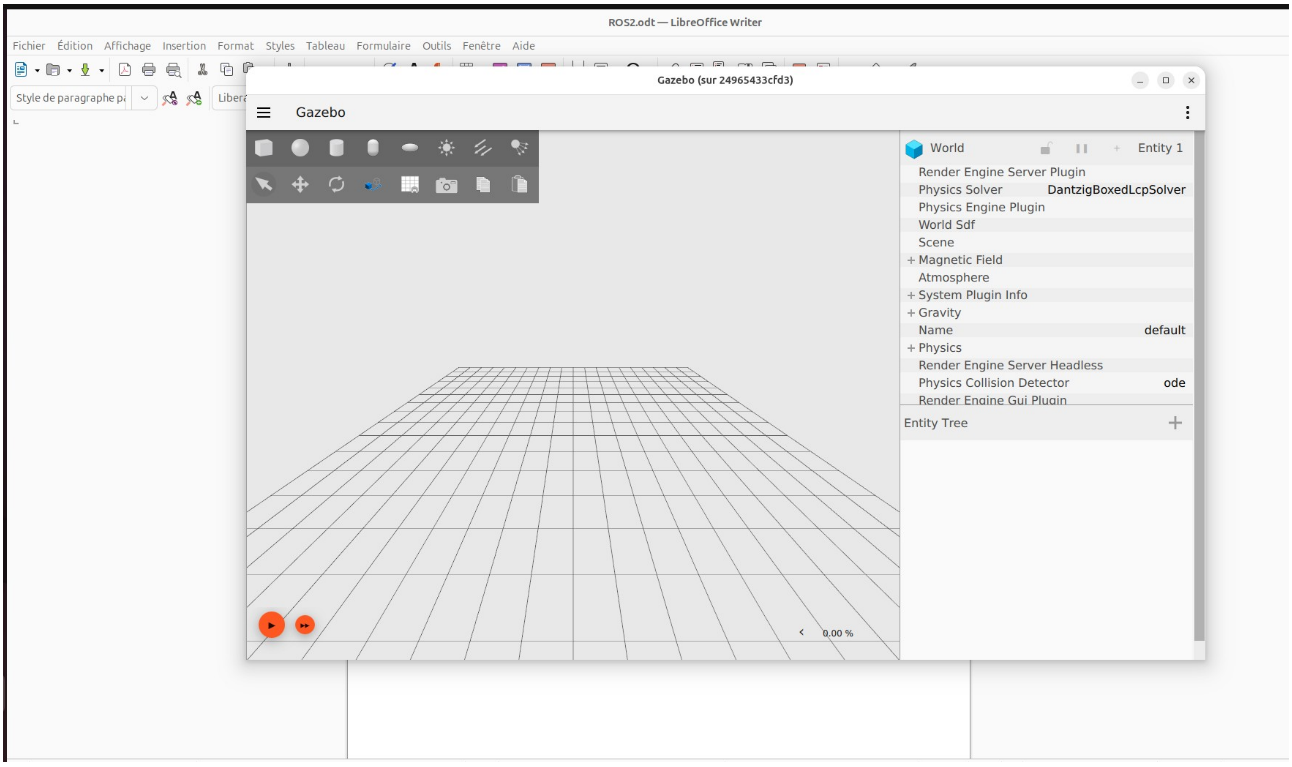
Introduction

In this part of the work, we will work on the thread 1 which is the design of a system allowing the autonomous navigation of the robot in the picking path, and to achieve this goal we will use ROS2.

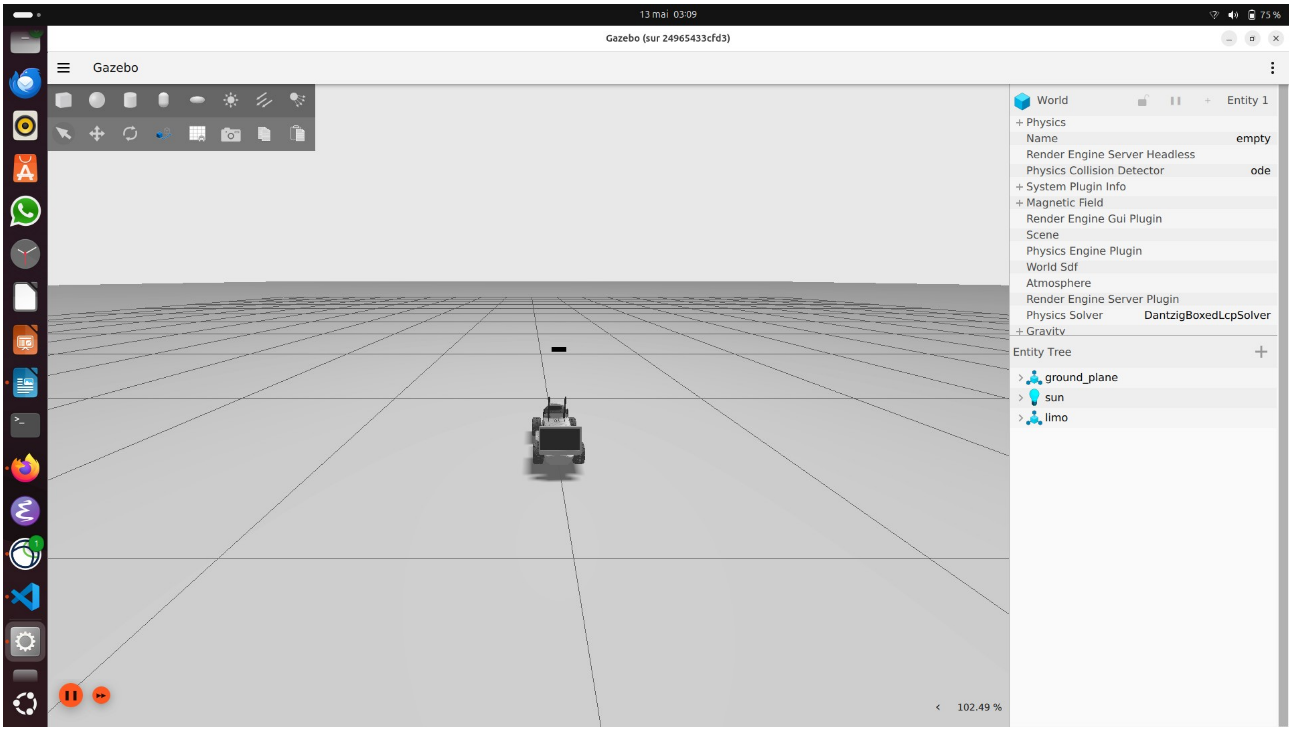
We planed to use Gazebo for the simulation of the robot but my computer doesn’t run Gazebo correctly, Gazebo is slow and the optimisation to make it rapid makes the car disappear. But Gazebo is not mandatory for the project, it is true that it can be useful, but we can use the ressources ROS2 gives for the robot simulation.

II-Gazebo installation :

After the configuration and the installation of the ressources related to Gazebo, I got the following window of Gazebo.



Result of the first simulation :



There are two problem with the simulation, the first one is that the car is Limo 01 instead of 02, and the second one is that the simulation is very slow on my computer, the alternative was to use the Lab computer but I found out that the Lab has only screens not the central processing unit. Finally, I plan to use the ressources given by ROS2 and give up Gazebo, like Rviz.

III-Get starting with ROS2

A- How to run ROS2

Here are the steps to runs ROS2 and use it :

On host terminal (Ubuntu 25.04) :

1- Allow the screen acces for Gzebo

xhost +local:docker

2- Checking if containers exist :

docker ps -a

3- Start the container :

docker start limo_navigation

4- verify if the container is running :

docker ps

In the container :

1- enter the container :

docker exec -it limo_navigation bash

2- sourcing the container :

source /opt/ros/humble/setup.bash

Listener :

- To listen :

ros2 topic echo /chatter std_msgs/msg/String

Subscriber :

-To send the data :

ros2 topic pub --rate 1 /chatter std_msgs/msg/String "{data: 'LIMO robot is ready'}"

B- Running executables